

Mathematics I

Week 5

Graded Assignment Practice Session

Plan for this session

- How to join?
 - Join on webex - click on link sent to you
 - Join on pear deck - joinpd.com (enter code: see top right on webex)
 - Keep a notebook and pen ready for solving problems
- For every question - 5 to 10 minutes allotted
 - Question will be shown on the screen for solving
 - If you are done solving, then enter your answer on joinpd.com
 - Instructor will provide the solution
 - Questions and discussion

Example Screenshots

Laptop/Desktop

The screenshot shows a presentation slide with a dark background. On the left, there is a large grey rectangular area. To its right, the text reads: "Is this function even or odd or neither even nor odd?". In the top right corner, a small box contains the text "How to participate? joinpd.com code". In the bottom left corner, there is a small logo with the letters A, B, and C, and the text "Students choose an option". In the bottom right corner, it says "Pear Deck Interactive Slide Do not remove this bar".

Q1 (a)

How to participate?
joinpd.com
code

Is this function even or odd or neither even nor odd?

Students choose an option

Pear Deck Interactive Slide
Do not remove this bar

Portion for Answering

☐ Even

☐ Odd

☐ Neither even nor odd

Mobile

The screenshot shows the same presentation slide as the laptop/desktop version, but adapted for a mobile phone screen. The top status bar shows the time as 2:50 PM and battery level at 63%. The address bar shows the URL "app.peardeck.com/studi". The slide content is the same as the desktop version. A red circle highlights the "Answer Question" button at the bottom right of the slide.

2:50 PM 63%

app.peardeck.com/studi

Q3

How to participate?
joinpd.com
code

A chemical substance A is the reactant in a chemical reaction which gets converted into a product B. The concentrations (in mol/L) of A and B depend on the reaction time t (min) as $C_A(t) = 20 - 2t^2 - 4t + 50$, $C_B(t) = 2t^2 + 2t + 44$.

How much time (in min) elapses after the reaction starts before the concentrations of A and B become equal?

Students, enter a number!

Pear Deck Interactive Slide
Do not remove this bar

Portion for Answering

Answer Question

Test Problem

How to participate?
joinpd.com
code: see above

This is a problem to test if everything is working.

How many marks do you expect in quiz-1?



Students, write your response!

Q1

Suppose $2p = q$, where $p, q \in \mathbb{R}$.

Q1

Suppose $2p = q$, where $p, q \in \mathbb{R}$. Find the value of $\frac{4^p}{2^q} + \frac{25^q}{625^p}$



Students, enter a number!

Solution

- $$\begin{aligned}\frac{4^p}{2^q} + \frac{25^q}{625^p} &= \frac{2^{2p}}{2^q} + \frac{25^q}{25^{2p}} \\ &= 2^{2p-q} + 25^{q-2p}\end{aligned}$$

Solution

- $$\frac{4^p}{2^q} + \frac{25^q}{625^p} = \frac{2^{2p}}{2^q} + \frac{25^q}{25^{2p}}$$
$$= 2^{2p-q} + 25^{q-2p}$$
- Given, $2p = q \Rightarrow 2p - q = 0$ (or) $q - 2p = 0$
substituting these values, we get

Solution

- $$\frac{4^p}{2^q} + \frac{25^q}{625^p} = \frac{2^{2p}}{2^q} + \frac{25^q}{25^{2p}}$$
$$= 2^{2p-q} + 25^{q-2p}$$
- Given, $2p = q \Rightarrow 2p - q = 0$ (or) $q - 2p = 0$
substituting these values, we get
- $2^{2p-q} + 25^{q-2p} = 2^0 + 25^0 = 1 + 1 = 2$

Solution

- $$\frac{4^p}{2^q} + \frac{25^q}{625^p} = \frac{2^{2p}}{2^q} + \frac{25^q}{25^{2p}}$$
$$= 2^{2p-q} + 25^{q-2p}$$
- Given, $2p = q \Rightarrow 2p - q = 0$ (or) $q - 2p = 0$
substituting these values, we get
- $2^{2p-q} + 25^{q-2p} = 2^0 + 25^0 = 1 + 1 = 2$

Answer: 2

Q2

Half-life of an element is the time required for half of a given sample of radioactive element to change to another element. The rate of change of concentration is calculated by the formula $A(t) = A_o(\frac{1}{2})^{(\frac{t}{\gamma})}$ where γ is the half-life of the material, A_o is the initial concentration of the radioactive element in the given sample, $A(t)$ is the concentration of the radioactive element in the sample after time t .

Q2

Half-life of an element is the time required for half of a given sample of radioactive element to change to another element. The rate of change of concentration is calculated by the formula $A(t) = A_o(\frac{1}{2})^{(\frac{t}{\gamma})}$ where γ is the half-life of the material, A_o is the initial concentration of the radioactive element in the given sample, $A(t)$ is the concentration of the radioactive element in the sample after time t .

If Radium has a half-life of 1600 years and the initial concentration of Radium in a sample was 100%, then calculate the percentage of Radium in that sample after 2000 years.



Students, write your response!

Solution

Step 1:

Given $A(t) = A_o(\frac{1}{2})^{(\frac{t}{\gamma})}$ and half life of radium is 1600 i.e $\gamma = 1600$

$$\begin{aligned}\Rightarrow A(2000) &= A_o(\frac{1}{2})^{(\frac{2000}{1600})} \\ &= A_o(\frac{1}{2})^{(\frac{5}{4})}\end{aligned}$$

Solution

Step 2:

At initial time($t = 0$) the concentration of Radium is 100%
 $\implies$ At initial time the concentration of Radium is A_o

Solution

Step 2:

At initial time($t = 0$) the concentration of Radium is 100%
 $\Rightarrow$ At initial time the concentration of Radium is A_0

Step-3:

So, the percentage of Radium in that sample after 2000 years is

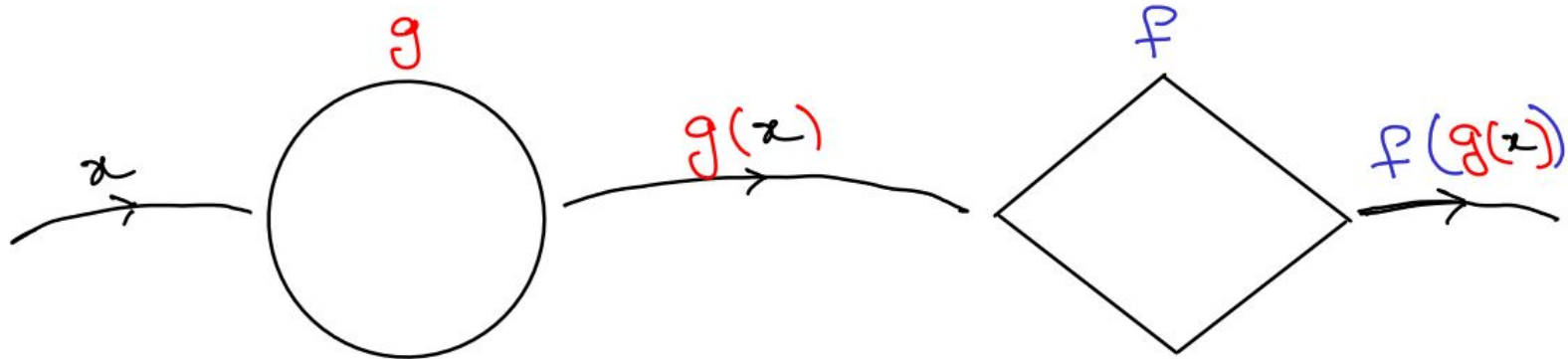
$$\left(\frac{A_0\left(\frac{1}{2}\right)^{\left(\frac{5}{4}\right)}}{A_0} \times 100\right)\% = \left(\left(\frac{1}{2}\right)^{\left(\frac{5}{4}\right)} \times 100\right)\% \approx 42\%$$

Answer: 42

Domain of the composite function $(f \circ g)$:

The domain of the composite function $f \circ g$ is the set of all 'x' such that

- 'x' is in the domain of g.
- $g(x)$ is in the domain of f.



$$(f \circ g)(x) = f(g(x))$$

Q3

Two functions $f(x) = (x + 1)^{\frac{1}{2}}$ and $g(x) = (x^2 - 1)$ are well defined.

Q3

Two functions $f(x) = (x + 1)^{\frac{1}{2}}$ and $g(x) = (x^2 - 1)$ are well defined. Find the domain of the composite function $g \circ f$.

- ☐ $\mathbb{R}$
- ☐ $((-3, \infty) \setminus ((-3, 5) \cap (-5, -1))$
- ☐ $[-1, \infty)$
- ☐ $\mathbb{R} \setminus (1, \infty)$



Students, write your response!

Solution

Step-1: Find the set of all 'x' such that 'x' is in the domain of the function 'f'.

Given, $f(x) = (x + 1)^{\frac{1}{2}}$.

For $f(x)$ to be well defined, $x + 1 \geq 0 \implies x \geq -1$.

So, the domain of $f(x)$ is $[-1, \infty)$.

Solution

Step-1: Find the set of all 'x' such that 'x' is in the domain of the function 'f'.

Given, $f(x) = (x + 1)^{\frac{1}{2}}$.

For $f(x)$ to be well defined, $x + 1 \geq 0 \implies x \geq -1$.

So, the domain of $f(x)$ is $[-1, \infty)$.

Step-2: Find the range of $f(x)$ and the domain of the function 'g'.

Now, the range of $f(x)$ is $[0, \infty)$ and the domain of $g(x) = x^2 - 1$ is $\mathbb{R}$.

Solution

Step-3: As the range of $f(x)$ is subset of the domain of $g(x)$.

Hence, when we use the two conditions to determine the domain of $g \circ f$.
Here, the domain of $g \circ f =$ The domain of $f = [-1, \infty)$.

Solution

Step-3: As the range of $f(x)$ is subset of the domain of $g(x)$.

Hence, when we use the two conditions to determine the domain of $g \circ f$.
Here, the domain of $g \circ f =$ The domain of $f = [-1, \infty)$.

Step-4: Find all the options that represents the interval $[-1, \infty)$

Now, $(-3, \infty) \setminus ((-3, 5) \cap (-5, -1)) = (-3, \infty) \setminus (-3, -1) = [-1, \infty)$.

Answer: (b),(c)

Q4

In a survey, the population growth in an area can be predicted according to the equation $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ where d is the percentage growth rate of population per year and T is the time since the initial population count α_o was taken.

Q4

In a survey, the population growth in an area can be predicted according to the equation $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ where d is the percentage growth rate of population per year and T is the time since the initial population count α_o was taken.

If in 2015, the population of Adyar was 30,000 and the population growth rate is 4% per year, then what will be the approximate population of Adyar in 2020? ($T = 0$ corresponds to the year 2015, $T = 1$ corresponds to the year 2016 and so on..)



Students, write your response!

Solution

Step-1:

Given $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ and initial population count $\alpha_o = 30000$.
The population growth rate $d = 4\%$ per year and $T = 5$.

Solution

Step-1:

Given $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ and initial population count $\alpha_o = 30000$.
The population growth rate $d = 4\%$ per year and $T = 5$.

Step-2:

Hence, the approximate population of Adyar in 2020 is $\alpha(5) = 30000 \times (1 + \frac{4}{100})^5 \approx 36500$

Answer: **36500**

Q5

If $f(x) = x^2$ in the restricted domain $[0, \infty)$, then choose the points where the graphs of the functions $f(x)$ and $f^{-1}(x)$ intersect each other?

Q5

If $f(x) = x^2$ in the restricted domain $[0, \infty)$, then choose the points where the graphs of the functions $f(x)$ and $f^{-1}(x)$ intersect each other?

☐ (0,0)

☐ (1,1)

☐ (-1,-1)

☐ (2,4)



Students, write your response!

Solution:

Step-1: Find the inverse function $f^{-1}(x)$.

Let us consider a function $g(x) = x^{\frac{1}{2}}$.

Now, $f \circ g = (x^{\frac{1}{2}})^2 = |x| = x$ and $g \circ f = (x^2)^{\frac{1}{2}} = |x| = x$.
($|x| = x$ because restricted domain is $[0, \infty)$).

Hence, g is the inverse function of f .

So, $f^{-1}(x) = x^{\frac{1}{2}} = \sqrt{x}$.

Solution:

Step-2: Find the x-coordinates of the intersection points by equating f and f^{-1} .

To get intersection points

$$\begin{aligned}f &= f^{-1} \\ \Rightarrow x^2 &= x^{\frac{1}{2}} \\ \Rightarrow x^4 &= x \\ \Rightarrow x^4 - x &= 0 \\ \Rightarrow x(x^3 - 1) &= 0 \\ \Rightarrow x((x - 1)(x^2 + x + 1)) &= 0 \\ \Rightarrow x(x - 1)(x^2 + x + 1) &= 0\end{aligned}$$

No real roots

$$\therefore x = 0, 1$$

Solution:

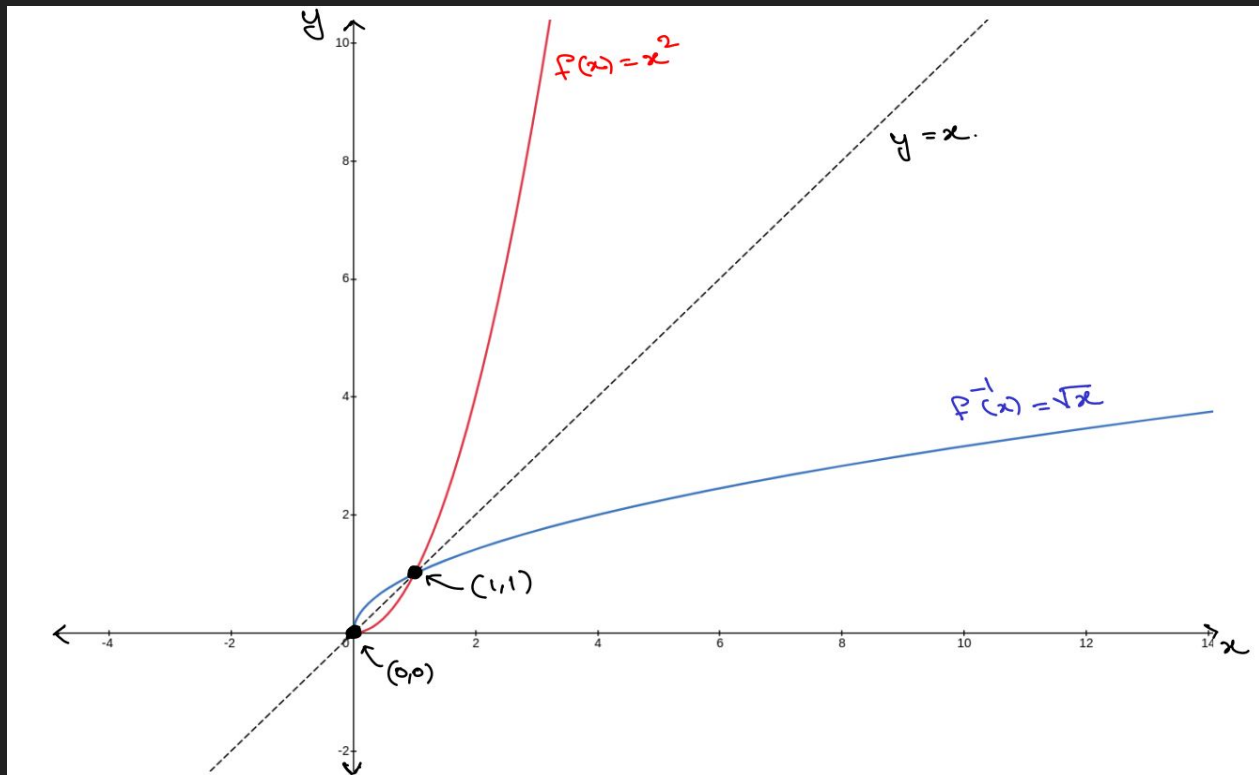
Step-3: Find the intersection points.

- The points are $(0, f(0))$ and $(1, f(1))$.
- $f(0) = 0$ and $f(1) = 1$
- Hence, the intersection points are $(0,0)$ and $(1,1)$.

Answer: (a) , (b)

Solution:

Alternate step: From the graphs of f and f^{-1} .



THANK YOU!!